

Airfuge™ Ultracentrifugation of Suspensions on Electron Microscopy Grids for Particle Enrichment

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----- Principles & Scope of Application -----

The procedure concentrates particles by ultracentrifugation directly onto a sample support for transmission electron microscopy (EM). For this purpose, a desktop air-driven ultracentrifuge from Beckman Coulter is used with the A-100/18 fixed-angle rotor and particular adaptors to hold the sample support, an EM grid.

Principle of the method: Particles of the sample suspension are forced towards the surface of an EM grid by a centrifugal acceleration of about 120.000 x g. Even the smallest viruses known are sedimented completely within 10 min at this acceleration.

Intended application: Enrichment of virus particles for diagnostic electron microscopy (reduction of the detection limit; Biel & Gelderblom 1999, Gelderblom et al. 1991, Laue & Bannert 2010).

Limitations: If the suspensions contain high amounts of unwanted particles, Airfuge enrichment results in overloading of the EM grids. See remarks for strategies avoiding overloading.

The procedure comprises several steps: (1) Pre-treatment of EM grids with Alcian Blue. (2) Placing the pre-treated EM grids into the recess of the adaptors. (3) Insertion of the adaptors into the vials filled with sample suspension. (4) Loading the rotor with the vials and ultracentrifugation. (5) Removal of the EM grids from the adaptors. (6) Negative staining of the EM grids.

----- Lab Safety -----

Safety rules differ locally. Please follow your local rules and/or consult your safety officer.

Chemicals can be harmful. Consider safety information, wear suitable personal protection (lab coat, gloves and goggles) and collect the waste according to local rules.

Samples may be harmful (e.g. infectious) or posing a risk to the environment. It is necessary to assess the risks of working with them. In most cases an official permission to work with infectious samples is necessary. Some general considerations regarding biosafety are summarized in the section "Remarks".

----- Materials -----

- (1) Sample suspension (at least 80 µl per vial; see also remarks)
- (2) Forceps for handling the grids and adaptors (inverted forceps are recommended)
- (3) Alcian Blue 8 GX (Sigma-Aldrich, Nr. 05500-5G)
- (4) Acetic acid, 100%
- (5) Negative stains: e.g. uranyl acetate, 1% in de-ionized water, pH 4.5, or phosphotungstic acid, 1% in de-ionized water, pH 7.
- (6) EM Grids, 400 mesh, copper, filmed with formvar or pioloform plastic and enhanced with a layer of carbon (~10 nm)
- (7) De-ionized water
- (8) Filter paper
- (9) Microliter pipette (10-100 µl) with single-use tips
- (10) Parafilm
- (11) Grid storage container (grid box, petri dish, etc.)
- (12) Cover for shielding the droplets with the grids during incubation
- (13) Desktop centrifuge
- (14) Vortex mixer
- (15) Magnetic stirrer
- (16) Ultrasonic bath (optional)
- (17) Airfuge (Beckmann Coulter) (Fig. 1)
- (18) A-100/18 Rotor, Fixed Angle, 6 x 175 µl, 95,000 rpm, 149,000 x g (Beckman Coulter, Part. No. 347593) (Fig. 2)
- (19) Polypropylene vials (tubes, 5x20 mm (Beckman Coulter, Part. No. 342630) (Fig. 3)
- (20) EM grid adaptor for Airfuge tubes (5x20 mm) (Beranek Laborgeräte; www.laborgeraete-beranek.de, Part. No. 11093) (Fig. 3)
- (21) Rotor caps, polyethylene (Beckman Coulter, Part. No. 339643) (Fig. 3)
- (22) Optional: Injection needle, bended to a hook.

----- Preparation of Alcian Blue -----

Stock Solution I:

- (23) Dissolve 2% (w/v) Alcian Blue in double distilled water. Put solution in an ultrasonic bath for 10 min to increase speed of dissolving.
- (24) Centrifuge solution for 1h at 10.000 – 14.000 x g at room temperature.
- (25) Collect supernatant carefully and store it at 4° C.

Stock solution II:

- (26) Mix acetic acid (2% v/v) with distilled water by slowly adding acid to the water while stirring the whole mixture (the mixture is getting extremely warm during mixing).

Solution for Application:

- (27) Mix Stock solution I with stock solution II in equal amounts. The final concentration of Alcian Blue is 1%.

The Alcian Blue solution can be stored for at least 6 months in the refrigerator (4-8° C).

As an **alternative**, a ready-made solution can be purchased from Sigma Aldrich (1.01647.0500). It can be diluted at 1:10 with de-ionized water.

----- **Procedure** -----

Six grids (= samples or suspensions) can be processed per Airfuge run in separate vials.

- (28) Centrifuge all solutions, except of the sample suspension, in a desktop centrifuge at highest speed for 10 min.
- (29) Fix a strip of parafilm on the desk.
- (30) Place a drop (30 µl) of Alcian Blue (1%) per sample on the parafilm.
- (31) Put the grids on the Alcian Blue droplets and incubate them for 1 min.
- (32) Vortex the sample suspension rigorously (three times for 10 sec each). Fill the vials with 80µl of the sample suspension.
- (33) Wash the EM grids by placing them briefly (1-2 sec.) on four droplets (30 µl) of de-ionized water. After the last washing step blot the grid using a filter paper and proceed immediately to the next step (do not dry the grid surface completely).
- (34) Dip the grid briefly into the sample suspension that is filled in the Airfuge vials and place it into the recess of the grid adaptor with the treated side facing upwards. Press the grid firmly to the bottom of the recess by using the tip of the forceps and moving around the rim of the EM grid.
- (35) **Alternative:** Insert the adaptors into the vial which is already filled with suspension (directly after step 32). Avoid the formation of air bubbles. Instead of step 34: Pull out the adaptor a little bit by using forceps or the bend injection needle and put the pre-treated EM grid into the recess of the adaptor, with the treated side facing upwards. Carefully re-insert the adaptors into the tubes completely.
- (36) Put the vials into the rotor with the grids facing towards the rotor axis (see Figs. 4). Cover the rotor with a plastic rotor cap (never run the Airfuge without rotor cap; Fig. 3).

- (37) Centrifuge at 20 psi (~85.000 rpm =~120.000 x g) for 10 min.
- (38) Add three drops (30 µl) of water and a single drop of the stain, either uranyl acetate or phosphotungstic acid, to the parafilm.
- (39) Pull out the adaptor from the tube using forceps or the bend injection needle and hold it with forceps.
- (40) Turn the adaptor that the grid faces downwards. Remove the loaded EM grid out of the adaptor's recess by adding a drop of double distilled water (~80 µl) on top of the adaptor (see Fig. 4). The release of the grid may last a few seconds. If the grid does not release by itself help it by touching the rim of the grid with the needle or forceps.
- (41) Wash the EM grid on the water droplets. Blot them and place the grids briefly (1-2 sec) on the negative stain.
- (42) Blot the grids dry with a filter paper (see remarks). Store the grids in a grid box container or petri dish.

----- **Remarks** -----

The standard **concentration of the staining solutions** is 1%. If dense or large particles in the sample are of interest (e.g. bacteria), reduction of the concentration to 0.5-0.2% may improve the visibility of structural details. As a standard for **diagnostic electron microscopy**, we prepare two samples of a suspension, one with either of the staining solutions.

For the generation of an appropriate **heavy metal contrast film**, the grid is blotted with a filter paper. **Blotting** the grid dry after phosphotungstic acid incubation usually results in sufficient staining quality (i.e. blotting is not critical). Uranyl acetate contrasting needs careful blotting otherwise the contrast film develops not sufficiently resulting in a "dry staining" which masks structural details. To produce an appropriate contrast film, touch the filter paper with the grid edge at an angle of about 45 ° and remove the grid before the fluid is completely absorbed by the filter paper. As an alternative, a wet filter paper can be used for blotting which reduces the suction force of the filter paper. Both procedures usually produce a contrast film with a gradual thickness on the grid which helps to find the perfect contrast for the particles of interest.

Forceps/rotor/adaptors should be cleaned directly after use by incubation in liquids. We use incubation in 5% Lysoform disinfection solution, which contains formaldehyde, for at least 10 min. After incubation we rinse the forceps/rotor/adaptors with de-ionized water and finally clean them with a paper towel soaked with 70% ethanol. It is possible to autoclave the **rotor** at 121°C for 30 min.

Biosafety rules are locally different. In any case, the operator should make a risk assessment for all procedures and should check if all legal aspects are met (e.g. whether an official permission is necessary to handle the sample or not). **Inactivation** of sample suspensions can be done by incubation with 2% paraformaldehyde in buffer (final concentration) for 30 min at 25 °C followed by 30 min at 37 °C on a shaker. This procedure inactivates even comparatively high concentrations of test viruses efficiently (Möller et al. 2015).

Pre-treatment of **clinical samples**, such as stool, swabs, blisters, tissue biopsies is described by Miller (1986). Larger and denser particles, which are unwanted can be

removed by low-speed centrifugation (e.g. 1000 x g for 2 min) prior to Airfuge centrifugation. **Filtering** using microporous filters (from 0.4 µm pore size upwards) is another possibility to remove larger particles. Suspensions with a high load of protein (e.g. serum) should be diluted 1/5 to 1/10 in de-ionized water prior to Airfuge centrifugation.

If the **sample volume** is smaller than 80 µl, the sample can be filled up with water or saline.

Increased number of **washing steps** after centrifugation may reduce the enrichment efficiency (Laue & Bannert 2010). As a standard, two washing steps are recommended. The release of the EM grid from the recess of the adaptors is already performed with water which already dilutes some of the non-bound material.

A **video** of the entire procedure is available at <https://doi.org/10.5281/zenodo.12705300> .

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----- **Figures** -----



Figure 1 Airfuge desktop ultracentrifuge



Figure 2 Airfuge fixed-angle rotor A-100/18 (without cap).

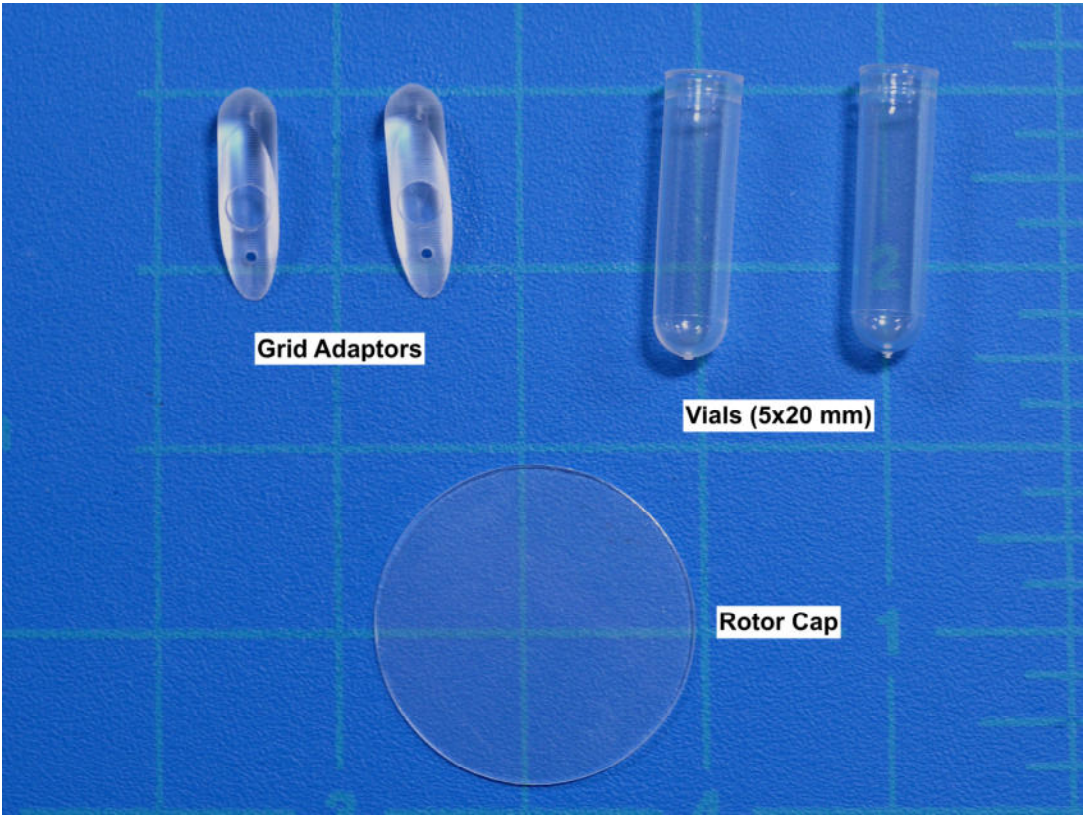


Figure 3 Grid adaptors, vials (5x20 mm) and rotor cap

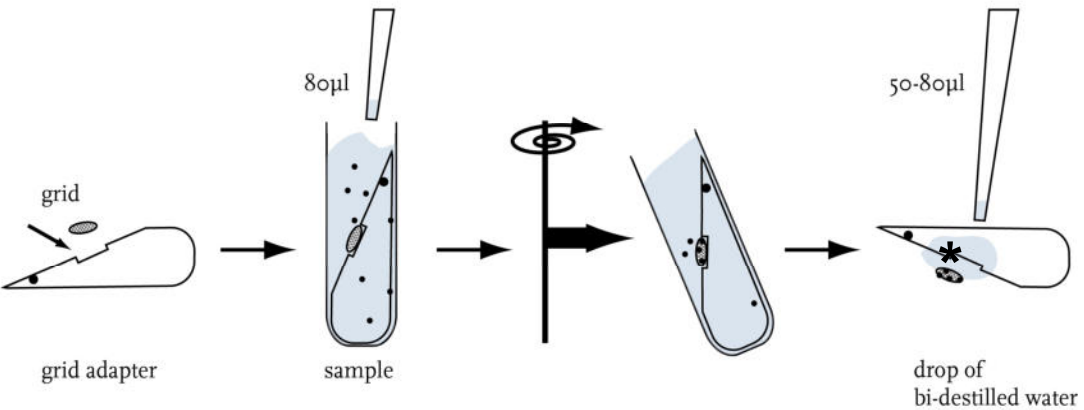


Figure 4 Assembly and disassembly of EM grids into/from grid-adaptors of the Airfuge

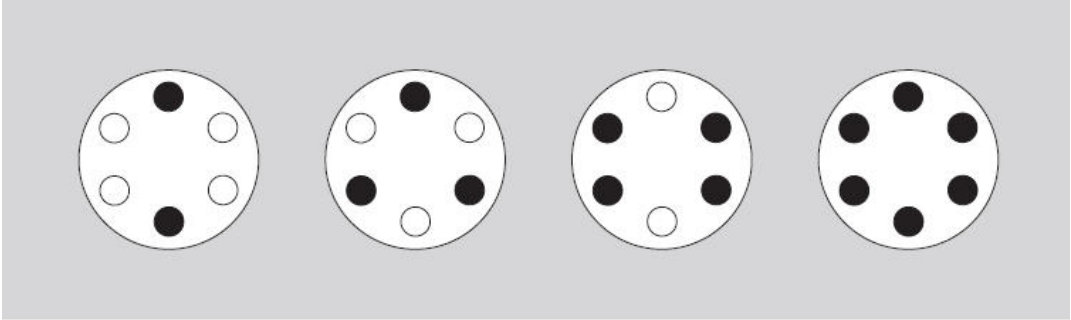


Figure 5 Possible positioning of vials (black) in the A-100/18 rotor. Source: <http://www.beckman.de/centrifugation/>

----- **Disclaimer** -----

The authors made every effort to provide timely and accurate information. Nevertheless, mistakes may occur. The author does not assume liability for relevance, accuracy and completeness of the information provided.

Please send comments/questions on the protocol directly to the author.